

VLM-Guided Keypoint Extraction for Scenario-Specific Sequential Tool Use

Süleyman Talha Belen
BSc EEE Student
Bilkent University
ID: 22202791

Yunus Emre Kök
BSc EEE Student
Bilkent University
ID: 22203752

Yusuf Buğra Yazıcıoğlu
BSc EEE Student
Bilkent University
ID: 22203518

Mehmet Emin Yakşı
BSc EEE Student
Bilkent University
ID: 22203637

I. PROBLEM STATEMENT

We address the problem of retrieving an object that is physically unreachable by a robot’s end-effector due to geometric constraints. To demonstrate generalizability, we formulate three distinct constrained scenarios: (1) retrieving an object from beneath a low-clearance shelf (requiring a pulling action), (2) extracting an object stuck inside a narrow horizontal pipe (requiring a pushing action), and (3) lifting a handled object from a deep vertical container (requiring precise vertical hooking).

Because a single Vision-Language Model (VLM) alone may not provide sufficiently accurate geometric estimates for precise manipulation, our system adopts a **dual-modality perception** approach. The **input** to our system is an RGB-D image of the scene. The **output** is a fully executed sequential manipulation plan. A VLM first performs high-level semantic reasoning to identify the appropriate tool and provide coarse pixel-level keypoint candidates for the tool’s grasp handle and functional tip. These candidates are then refined by a dedicated geometric keypoint separation and pose estimation module, which leverages depth data and point cloud fitting to produce precise 3D tool poses directly in the simulation coordinate frame. The resulting keypoints drive a classical trajectory optimization framework (KOMO) to execute the manipulation sequence in a simulated Franka Emika Panda environment (RAI).

II. MOTIVATION AND IMPACT

Tool use is an unavoidable part of intelligent behavior, yet it remains challenging due to the need for both semantic reasoning (which tool to use?) and precise geometric execution (how to move it?). By replacing generic objects with three task specific tools and distinct scenarios, this project evaluates a robot’s ability to generalize its tool use capabilities based on the environmental context.

A key motivation for adopting a dual-modality pipeline is robustness: while VLMs excel at semantic scene understanding, they cannot be relied upon alone for precise continuous keypoint localization. A geometric pose estimation module compensates for this limitation, ensuring that the trajectory optimizer receives accurate and stable 3D keypoints regardless of VLM uncertainty. This project directly addresses multiple

prioritized topics from the course: *sequential robotic manipulation*, *contact-rich manipulation*, and *deep perception for robotic manipulation*. If successful, it demonstrates harnessing the semantic strengths of foundation models while transferring rigorous geometric estimation and collision-free trajectory generation to classical methods like KOMO.

III. RELATED WORK

Our approach builds upon recent advances in combining foundation models with robotic control. SayCan [1] demonstrated how LLMs can be grounded in robotic affordances for high-level task planning. While SayCan focuses on discrete skill selection, we extend this by using a VLM for fine-grained semantic reasoning, complemented by a geometric pose estimation module for precise keypoint localization. VoxPoser [2] synthesizes dense 3D value maps from language models to guide trajectories. We adopt a more modular approach: the VLM provides coarse keypoint candidates, which are then refined geometrically before being consumed by a trajectory optimizer (KOMO). This design (combining high-level reasoning to learned models while relying on model based methods for precise motion generation) is a central theme in current manipulation research [3].

IV. PROPOSED APPROACH

Our pipeline consists of three main stages.

Scene Setup and Perception. We build three tabletop environments in RAI, each containing a target object, environmental constraints (shelf, arch, or deep container), and the three candidate tools (L-hook, pusher stick, J-hook). At the start of an episode, an RGB-D image is captured from the simulation camera.

VLM-Based Tool Selection and Coarse Keypoint Estimation. The RGB image is queried to a VLM with a prompt requesting: (a) identification of the appropriate tool for the presented obstacle, and (b) pixel-level keypoint coordinate candidates for the selected tool’s handle and functional tip. The VLM is only responsible for high-level semantic identification and providing coarse spatial estimates.

Geometric Keypoint Separation and Pose Estimation. Since VLM outputs alone may be insufficiently accurate for precise control, a second modality refines the coarse estimates. Using the depth channel of the RGB-D image, the tool’s

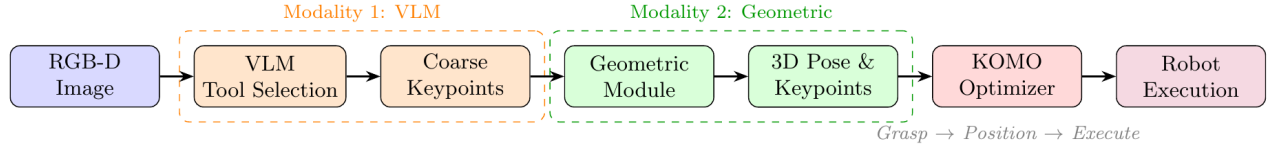
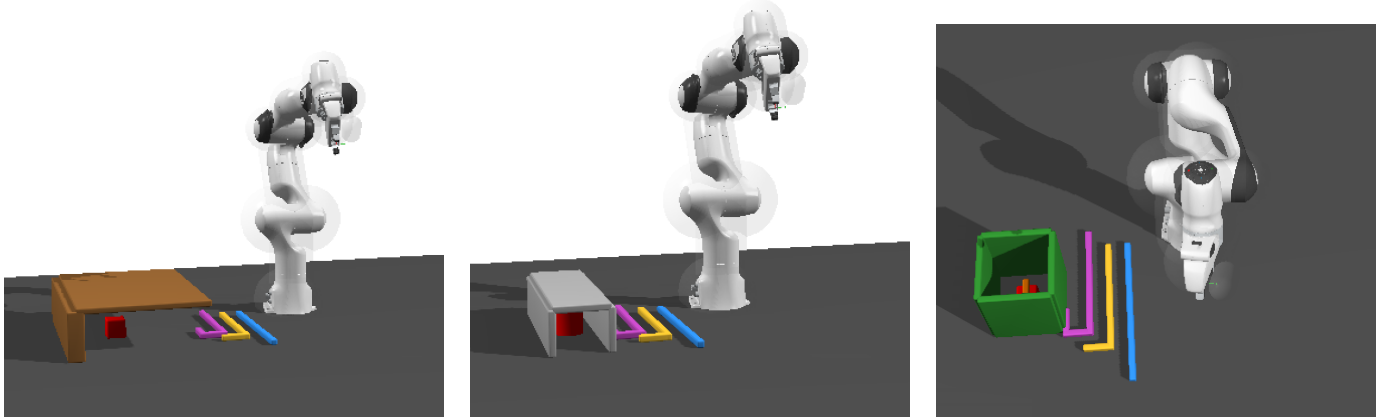


Fig. 1: Overview of the proposed dual-modality pipeline. The RGB-D image is first processed by the VLM for tool selection and coarse keypoint estimation (Modality 1), followed by geometric refinement for precise 3D pose estimation (Modality 2), and finally sequential trajectory optimization via KOMO.



(a) Scenario 1: Shelf (Pulling). A red block is trapped beneath a low-clearance wooden table. The robot should use the L-shaped hook to pull it out.

(b) Scenario 2: Arch (Pushing). A red block is trapped under a low-clearance arch structure. The robot selects the appropriate tool (blue, long stick) to retrieve it.

(c) Scenario 3: Deep Container (Hooking). A handled object sits inside a deep green bin. The robot uses the J-shaped hook to lift it out vertically.

Fig. 2: The three simulated RAI environments used in this project. Each scenario features the Franka Emika Panda robot alongside three candidate tools: an L-shaped hook (yellow), a straight long pusher stick (blue), and a J-shaped hook (magenta).

point cloud is extracted in the region suggested by the VLM. A geometric fitting procedure then separates the handle and functional tip keypoints and estimates the full 6-DoF pose of the selected tool directly in the simulation coordinate frame. These 3D keypoints are then passed to KOMO to drive a multi-phase manipulation sequence: (1) *Grasping*: the robot navigates to the 3D handle keypoint and closes the gripper; (2) *Positioning*: the tool is oriented appropriately for the specific scenario; (3) *Execution*: the robot uses the tool’s functional tip keypoint to push, pull, or hook the target object. Each phase features collision-avoidance and joint limit constraints.

V. RELEVANT COURSE TOPICS

This project covers: *forward and inverse kinematics*, *geometric and deep perception* (depth-based point cloud fitting, pose estimation, and VLM keypoint extraction), *motion planning and trajectory optimization* (KOMO), *sequential manipulation*, and *contact-rich manipulation*.

VI. PHASE GOALS

By Progress Report:

- Functional RAI simulation environments for all three scenarios.

- Working VLM pipeline for accurate tool selection and coarse 2D keypoint estimation.
- Functional geometric keypoint separation and pose estimation module.
- Successful basic tool grasping in simulation.

By Final Report:

- Full execution of the manipulation sequences (pull, push, hook) across all three environments.
- KOMO-based trajectory optimization with collision constraints.
- Quantitative evaluation of both VLM coarse keypoint accuracy and geometric refinement accuracy, as well as overall task success rate.

VII. TASK SHARE

Süleyman Talha Belen:

- VLM coarse keypoint extraction pipeline development
- Geometric keypoint separation and pose estimation module development.
- KOMO trajectory optimization formulation and sequential phase design (together with Yunus Emre).

Yunus Emre Kök:

- KOMO trajectory optimization implementation and collision-avoidance constraint design and tuning (together with Talha).
- Grasping and positioning phase implementation.

Yusuf Buğra Yazıcıoğlu:

- RAI simulation environment setup for all three scenarios.
- Tool geometry modeling (L-shaped hook, straight pusher stick, J-shaped hook).

Mehmet Emin Yakşi:

- Depth-based point cloud extraction and tool fitting pipeline.
- KOMO execution phase (push/pull/hook movements).
- Evaluation metrics design and task success rate measurement.

REFERENCES

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